

Question:

A 31-year-old man comes to the physician because of pain, tingling, and numbness in his right hand that started 3 months ago. It is worse at night and frequently wakes him up. The symptoms can be relieved by shaking his hands but soon recur. He reports weakness of his right hand, especially when grasping objects. He has type 2 diabetes mellitus. His current medications are metformin and sitagliptin. Four months ago he went on a camping trip. He has been working as a hardscaper for 8 years. His temperature is 37.5°C (99.5°F), pulse is 86/min, and blood pressure is 110/70 mm Hg. Examination shows reproduction of his symptoms when his right hand is held above his head for 2 minutes.

Laboratory studies show:

Hemoglobin 13.2 g/dL

Leukocyte count 7,600/mm³

Hemoglobin A1C 6.3%

Erythrocyte sedimentation rate 13 mm/h

Which of the following is most likely to confirm the diagnosis?"

Answer Choices:

- A. CT scan of cervical spine
- B. Nerve conduction studies
- C. MRI of the head
- D. Arterial Doppler ultrasonography

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Reasoning + Answer:

<think>

The patient is experiencing symptoms of carpal tunnel syndrome , including pain , tingling , and numbness in the right hand , which can be relieved by shaking the hands but recur quickly . Additionally , he has weakness in his right hand , especially when grasping objects , which is also consistent with carpal tunnel syndrome . The history of camping and his occupation as a hardscaper suggest repetitive strain and potential nerve compression .

</think>

<answer>

B . Nerve conduction studies

</answer>

